

Exercise 1

Using $E_\nu \approx 1 \text{ MeV}$, estimate the recoil energy and recoil velocity of B^*

Then, assuming the recoil loses directional memory after traveling a few angstroms in a solid, estimate τ_{stop}

Exercise 2

A neutrino produced by a left-chiral charged-current interaction is not a pure helicity eigenstate if $m_\nu \neq 0$.

- ★ Show that for $E_\nu \gg m_\nu$, the probability of the opposite-helicity component is approximately

$$P_{\text{opp}} \simeq \left(\frac{m_\nu}{2E_\nu} \right)^2.$$

- ★ Using $E_\nu = 1 \text{ MeV}$ and $m_\nu < 0.45 \text{ eV}$, estimate P_{opp}
- ★ Explain why this makes helicity and chirality effectively equivalent for relativistic neutrinos.